IQP Generative Models for Strong Gravitational Lensing Images

Anonymous Author(s)
Affiliation
Address
email

Abstract

Instantaneous-quantum-polynomial (IQP) circuits are believed hard to sample classically [4, 5], while the expectation values a maximum-mean-discrepancy (MMD) loss needs stay computable, so such a Born machine trains classically and deploys on quantum hardware [15]. A recent correlation–complexity map, two indices from a dataset's Walsh order spectrum and its multi-information, predicts which datasets suit them, and DeepLense strong-lensing images sit at the most IQP-compatible position we measured (18-bit tuples 0.47, 0.87; the 10-bit crop pmf we train on 0.47, 0.43), above the turbulence data it was introduced on. Given *learned* latent directions $\theta = \theta^0 + \Phi^\top z$, a 10-qubit machine on lens crops is competitive with, and on some metrics matches, an RBM and a DCGAN carrying 2–39$\times$ more parameters at 11 training images, and reconstructs unseen images at correlation 0.9684 ± 0.0004. Three caveats bound it. The reference protocol's fixed latent block yields only noise, a structure-matched permutation null still scores $I_{\mathrm{QCLI}} = 0.42$, and the DCGAN leads at 100 images, as the reference paper predicts. Everything is exact classical simulation at $n = 10$ qubits over five seeds, so we claim no quantum advantage.

1 Introduction

IQP circuits are commuting-gate circuits whose output distribution is the squared Walsh–Hadamard transform of a phase function. Exact classical sampling from them would collapse the polynomial hierarchy [4] and the hardness is conjectured to survive approximate sampling [5], though sufficient noise restores classical simulability [14] and classical algorithms exist for some IQP instances [7]. The same commuting structure keeps an MMD loss's expectation values computable, so such a model can be *trained classically and deployed on quantum hardware* [15, 16].

Whether real data sits where that is useful is an empirical question, and a recent proposal [11] approaches it with two ingredients. (i) A *correlation–complexity map* placing a dataset on two coordinates, the quantum correlation-likeness indicator I_{QCLI}, the Jensen–Shannon divergence between the data's Walsh order spectrum and the i.i.d. binomial baseline $b_k = \binom{n}{k}/2^n$, and the classical correlation–complexity indicator I_{CCI}, how much of its total correlation the best pairwise tree misses [19, 17, 6], the high-I_{QCLI}, high-I_{CCI} corner being where IQP circuits should do well. (ii) A latent-adaptation protocol (Algorithm 1 of the reference paper, which trains a shared phase core on one anchor, adapts a fixed block of 50 Walsh-basis phases in Hamming order sample by sample and interpolates it to unseen samples), both demonstrated on 2D turbulence snapshots [10] at $n = 18$ qubits.

We take both to simulated dark-matter strong-lensing images from DeepLense [1, 3]. Lens crops land at the map's most IQP-compatible position, a 10-qubit IQP Born machine trained on them is competitive with the paper's own RBM and DCGAN baselines on both metrics at 2–39$\times$ fewer

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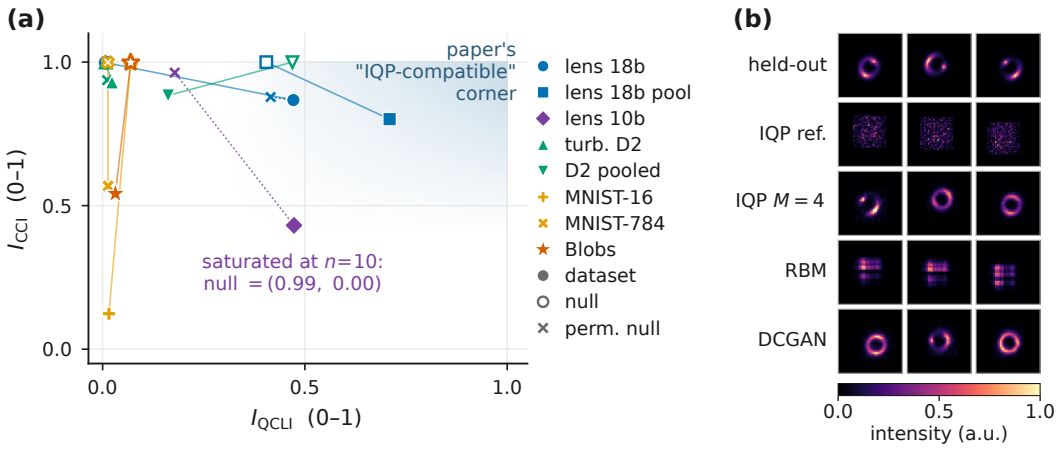


Figure 1: **(a)** Correlation–complexity map, datasets filled, i.i.d.-uniform nulls open and labelled null, permutation nulls as crosses (ours, not in the paper). In the legend lens 18b and 18b pool are the 18-bit per-image and pooled encodings, lens 10b the 10-bit crop pmf, turb. D2 and D2 pooled the paper's turbulence data, MNIST-16 and MNIST-784 MNIST at 4×4 and 28×28, Blobs BinaryBlobs. **(b)** First three unconditional samples per model, N=100, no selection, shared 0–1 colour scale.

parameters at 11 images, and reaches that level only once the protocol's fixed Walsh latent block is made *learnable*. Appendices B–D take the same tests back to the paper's own data, where its turbulence result does not reproduce under our pre-registered convention and two internal claims come out mixed.

2 Setup

Model. We compare the reference protocol against its fixed 50-phase block trained *jointly* across images and against our *learned-direction* model, a trainable Φ of M phase-space directions, each image a point z_i in M dimensions,

$$\theta_i \;=\; \theta^0 + \Phi^\top z_i, \qquad \theta^0 \in \mathbb{R}^{1024}, \quad \Phi \in \mathbb{R}^{M\times 1024}, \quad z_i \in \mathbb{R}^M, \tag{1}$$

with M=4 in Table 2 and $M \in \{0, 2, 4, 6, 8, 16\}$ in the Fig. 2(a) sweep, each row of Φ L^2-normalised to $\sqrt{1024}$, and a relative L^2 penalty on z of $\lambda = 10^{-3}L_0$, raised to $0.3L_0$ for M=8, L_0 being the data loss at θ^0 and $z = 0$. The objectives are the paper kernel (the multi-bandwidth Gaussian MMD kernel the reference paper takes from IQPopt [16], whose bandwidth rule we follow since the paper gives none, App. E), the flat MMD (uniform Walsh weights, estimable from hardware shots) and exact KL (simulation-only); z is drawn from a fitted Gaussian or resampled with jitter, and θ^0 is warm-started by 1,000 exact-KL steps, itself simulation-only.

Baselines and metrics. The baselines are a restricted Boltzmann machine (RBM; 10 visible and 1,000 hidden units, persistent contrastive divergence with one Gibbs step [9, 18]) and a deep convolutional generative adversarial network (DCGAN) [13], sized in Table 2, both early-stopped on 200 validation images disjoint from the held-out set (160k or 80k steps for the RBM, 5k to 30k for the DCGAN) while the IQP arms run a fixed 4,000, which can only favour them. We keep the reference paper's two metrics, the Jensen–Shannon divergence between intensity histograms (PDF–JS) and Conv2D-MMD^2, a biased V-statistic in a fixed random convolutional encoder's feature space [8], and add a held-out projection correlation. PDF–JS uses 50 bins fixed on $[0, 1]$ and the Conv2D-MMD^2 scale and bandwidth come from the held-out set alone, so neither metric moves with the arm scored, and the floor is 20 draws of 200 against 200 disjoint validation images (App. E). Every IQP arm and the RBM share the flux scalar S while the DCGAN learns its own, so raw Conv2D-MMD^2 is partly a brightness comparison and we add a flux-aligned variant, flagged as post-hoc, rescaling every image to the held-out mean total intensity. Every arm generates 200 images scored against the same 200 held-out images at $N \in \{11, 100\}$, every number a mean $\pm$ sample sd over five seeds, and the projection optimises z from eight restarts (App. E).

Table 1: Correlation–complexity map on our data, each permutation null keeping the value histogram of the row above, mean ± sd over five seeds. K is the number of n-bit codes per image or snapshot; the lens 18-bit row sweeps all $4{,}096$ (k_x, k_y) cells of a 64×64 image and so has no sampling noise, while the turbulence row subsamples a larger grid. Both are K-dependent, so a row reads only against its own-K null, and at the paper's own $K = 5 \times 10^4$ the turbulence I_{QCLI} is 0.091 ± 0.000 against a null of 0.098 ± 0.001.

Dataset and encoding	K	I_{QCLI}	I_{CCI}
Lens, 18-bit $(x, y, \mathrm{intensity})$ tuples	4,096	0.4715 ± 0.0000	0.8677 ± 0.0000
permutation null	4,096	0.4154 ± 0.0000	0.8783 ± 0.0001
Turbulence D2, 18-bit tuples	4,096	0.0232 ± 0.0002	0.9285 ± 0.0008
permutation null	4,096	0.0114 ± 0.0000	0.9371 ± 0.0009
Lens, 10-bit crop pmf	exact pmf	0.4731 ± 0.0000	0.4313 ± 0.0000
permutation null	exact pmf	0.1782 ± 0.0000	0.9634 ± 0.0004

Departures from the reference setup. Six choices differ from the reference paper, each following from the data. Its snapshots interpolate to unseen times, whereas the DeepLense `Model II` images we use [1, 2, 12] are static in three classes, so we generate unconditionally from the latent prior. It encodes a sample as 18-bit $(x, y, \mathrm{intensity})$ tuples, whereas we centroid-align each 64×64 image and read its central 32×32 crop, 99.4% of the light, as one probability mass function (pmf) over $1{,}024$ cells at $n = 10$ (App. A), a raw intensity axis carrying barely one bit. Its anchor-and-adapt protocol fixes the latent block, whereas we learn Φ and train the N images jointly (Sec. 3.2), and it estimates its MMD from 500 samples per step where a 10-qubit pmf is exact. It holds out times, which static images lack, so we project onto held-out images. Each generated crop is pasted back into 64×64 at the training-set mean total intensity S (112.3 at N=11, 124.9 at N=100; held-out 123.9) and a training-set offset.

3 Results

3.1 Lens images are the most IQP-compatible dataset on the map

Table 1 recomputes the map. Encoded as the reference paper encodes turbulence, lens images sit far above it in I_{QCLI} and inside the corner the map calls IQP-compatible, yet a permutation null that keeps each image's intensity histogram and destroys spatial structure nearly matches them, and turbulence behaves the same way, so at this encoding the map largely tracks the histogram. Only the 10-bit crop pmf we train on separates from its null in both coordinates, so a favourable position is necessary, not sufficient.

3.2 Why the reference protocol needs learned directions

Run sequentially on our low-variance latent block, Algorithm 1 produces noise (Fig. 1(b), second row; App. E has the Hamming-ordered variant). Representability is not the cause, since the full 1,024-phase circuit fits a *single* aligned crop to correlation 0.99997 ± 0.00001 under the flat MMD. The protocol is. With the core frozen on one anchor, the 50 fixed latent coordinates reconstruct every other training image at only 0.44 ± 0.04, fixed Walsh directions cross-fading pixels as an XOR-shift interference pattern instead of moving lens structure.

3.3 Learned latent directions generalise

Equation (1) repairs this. Freezing θ^0 and Φ and optimising only z on 200 *held-out* images, the best arm at N=100, flat MMD at M=4, reaches mean correlation 0.9684 ± 0.0004 against a nearest-training-image reference of 0.970, the paper kernel reaching only 0.792 ± 0.006 at N=11 and 0.805 ± 0.012 at N=100. The gain comes from learning the directions, not the extra parameters, since in the N=20 sweep of Fig. 2(a) a trained Φ beats a random direction on the same latent-free offset (0.7458 ± 0.0005). Useful M also stays small, M=2 being the best and by far the most stable (0.9060 ± 0.0021 against 0.8577 ± 0.0559 at M=4), with collapse by M=8 (0.218 ± 0.121) once the required $\|z\|$ exceeds the coherence radius.

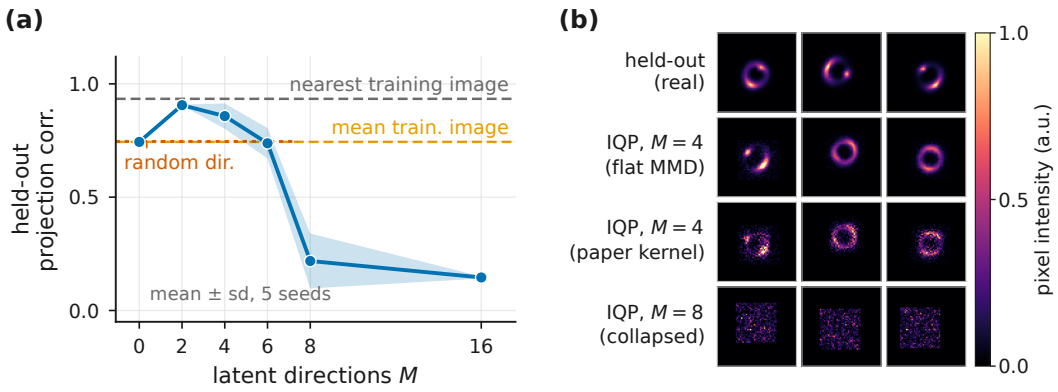


Figure 2: Different experiments. **(a)** Controlled latent-width sweep (N=20), held-out projection correlation against M, mean ± sd over five model seeds, with its own reference levels (0.934 nearest training image, 0.744 mean training image, 0.746 a random direction on the latent-free M=0 offset); M=8 and 16 are best over a learning-rate sweep selected on the plotted correlation, M=0 and M=4 single-rate. **(b)** Round-3b arms at N=100, held-out crops, then M=4 flat MMD, M=4 paper kernel and the collapsed M=8 arm.

Table 2: Paper-style comparison on lens crops, mean ± sd over five seeds, PDF–JS in 10^{-3} and both MMD^2 columns in 10^{-2}. Lower is better; **bold** marks a best mean leading the runner-up by more than both sds, so a column without bold carries no win the seed spread supports. Params are totals at N=11 / N=100; MMD$^2_{\text{al}}$ is flux-aligned (post-hoc) and the last row a real-vs-real floor over 20 draws of 200 against 200. App. E gives † and the parameter convention.

		$N = 11$			$N = 100$		
Model	Params	PDF–JS	MMD^2	MMD$^2_{\text{al}}$	PDF–JS	MMD^2	MMD$^2_{\text{al}}$
IQP, reference protocol (Alg. 1)†	1,524 / 5,974	7.2 ± 0.7	18.8 ± 1	22.5 ± 2	8.4 ± 0.4	21.8 ± 0.6	26.6 ± 0.7
IQP, fixed basis, joint	1,524 / 5,974	1.6 ± 0.4	12.3 ± 0.4	12.5 ± 1	1.0 ± 0.1	10.2 ± 0.8	8.5 ± 1
IQP, learned dir. M=4 (flat MMD)	5,164 / 5,520	3.0 ± 0.2	8.9 ± 0.2	2.7 ± 0.2	2.9 ± 0.3	7.2 ± 0.3	2.6 ± 0.4
IQP, learned dir. M=4 (paper kernel)	5,164 / 5,520	0.81 ± 0.05	9.6 ± 0.3	7.4 ± 0.6	1.4 ± 0.2	11.3 ± 0.5	10.0 ± 0.8
RBM [18]	11,510 / 15,960	1.7 ± 0.8	6.5 ± 0.8	11.1 ± 4	0.9 ± 0.1	5.9 ± 0.4	8.1 ± 2
DCGAN [13]	199,837	1.1 ± 0.6	5.1 ± 2	3.9 ± 1	**0.55** ± 0.12	**2.2** ± 0.3	2.1 ± 0.4
Real-vs-real floor	n/a	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2	*0.04* ± 0.02	*0.5* ± 0.3	*0.4* ± 0.2

3.4 Paper-style comparison and data scaling

In Table 2 the eleven-image regime is where the IQP arms are competitive. The M=4 paper-kernel arm holds the best mean PDF–JS there and the M=4 flat-MMD arm the best mean flux-aligned Conv2D-MMD^2 at 2–39× fewer parameters, but both leads sit inside one sd of the baselines, so we read them as matching, not beating. Raw Conv2D-MMD^2 runs the other way at both sizes, both baselines ahead of every IQP arm, an ordering the flux-aligned column partly reverses. At N=100 the DCGAN leads all three columns, decisively on PDF–JS and raw Conv2D-MMD^2 and within the M=4 flat-MMD arm's spread on the flux-aligned one. That ordering is the reference paper's data-scaling picture, traced across N in Fig. 5. Trained inside one class, the same arm holds a projection correlation of 0.969–0.972 with PDF–JS and Conv2D-MMD^2 at the pooled level, each against its own floor (Table 4).

4 Discussion and limitations

Nothing here is evidence of quantum advantage, since every circuit is simulated exactly at $n = 10$ qubits and the hardness results describe the class at scale, not our instances. Five seeds are few, and the DCGAN adds a spread of up to 57% on raw Conv2D-MMD^2 and 74% on PDF–JS at one configuration and seed, cuDNN having no deterministic transposed-convolution kernel. S is imposed on the IQP arms and the RBM but learned by the DCGAN.

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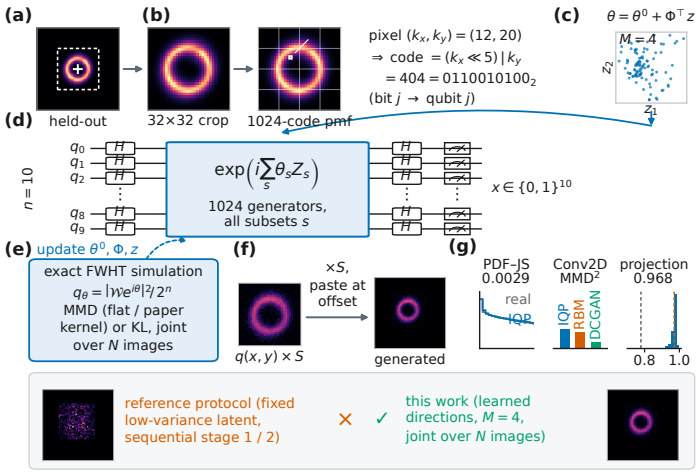


Figure 3: Pipeline. **(a)** A held-out 64×64 image, white cross at its intensity centroid and dashed box the 32×32 crop window. **(b)** The aligned crop read as a pmf over 1,024 cells, the example pixel $(k_x, k_y) = (12, 20)$, the integer form of (x, y), carrying code $404 = 0110010100_2$ with bit j addressing qubit j. **(c)** The first two learned latent coordinates of the 100 training images at $M = 4$. **(d)** The IQP circuit, an H layer, one diagonal phase $\exp(i \sum_s \theta_s Z_s)$ over all 1,024 subsets s, a second H layer and measurement. **(e)** Exact simulation through $q_\theta = |\mathcal{W} e^{i\theta}|^2 / 2^n$, trained jointly over the N images by MMD (flat or paper kernel) or by KL, the fast Walsh–Hadamard transform (FWHT) supplying $\mathcal{W}$ and the dashed arrow updating θ^0, Φ and z. **(f)** A trained $q(x, y)$ scaled by S and pasted at a sampled offset. **(g)** The intensity PDF of real against IQP samples (PDF–JS 0.0029), Conv2D-MMD^2 at N=100 (IQP 0.072, RBM 0.059, DCGAN 0.022) and projection correlations over 200 held-out images (mean 0.968). The bottom row contrasts a reference-protocol sample with an M=4 sample from this work.

A Pipeline

Figure 3 summarises the encoding, the latent parameterisation and the evaluation.

B Replication of the reference paper's turbulence experiment

We attempted a direct replication on the reference paper's own dataset (2D cylinder-wake turbulence D2 [10], $n = 18$ qubits, 11 anchor snapshots) over five seeds. Under our pre-registered convention the result is **not reproduced**, since in the 11-snapshot regime the RBM (PDF–JS 0.217 ± 0.007) and a 1.2M-parameter DCGAN (0.045 ± 0.062) beat our 31k-parameter IQP (0.272 ± 0.004) on *both* metrics. Two findings qualify this. Under a last-write-wins aggregation convention, never specified in the paper, our IQP's PDF–JS values land close to the published ones (0.0324 ± 0.0011 and 0.0473 ± 0.0005 against 0.0238 and 0.0221, with Conv2D-MMD^2 0.711 ± 0.012 and 0.738 ± 0.017), but the RBM still wins. Conv2D-MMD^2 is also saturated, ground truth shuffled in pixel space already scoring 0.7045 ± 0.0095, below every model in the paper's Table 2 (0.90–1.47), and unseen-time synthesis is solvable by linear interpolation (0.9998, deterministic). Our anchor-fit level (0.651 ± 0.010 sampled, 0.739 ± 0.010 exact for the 31k arm) traces to stage 1 freezing the latent block (-0.114 ± 0.021) and to a 0.892 model-family ceiling that carries no seed, nothing in that fit consuming the random stream, rather than to our implementation; the 5×10^4-sample pmf, which a single seed made look like a second cause, costs nothing measurable over five (-0.001 ± 0.022), and stage 2 adds 0.000 ± 0.000.

C Support mismatch (reference paper Fig. 3)

The reference paper's Fig. 3 claims (a) that the upper envelope of achievable MMD falls as the target's I_{QCLI} rises, because the support-induced floor shrinks, and (b) that the envelope drops overall as the learner gets more gates (Fig. 4(a)). Claim (b) reproduces, the median achieved MMD^2 falling $9.58 \times 10^{-3} \to 2.15 \times 10^{-3} \to 1.44 \times 10^{-3}$ across the three learners with a seed spread of at most



Figure 4: **(a)** Support-mismatch study at $n = 16$, with 140 target circuits ($\leq$ 4-local) spanning exact $I_{\mathrm{QCLI}} \in [0, 0.408]$ fitted by $\leq$ 2-local learners of $G_n = 50$, 100 and 150 requested gates, G_n being the learner's gate count, the last capped at the 136 generators a $\leq$ 2-local circuit has at $n = 16$ (exact MMD, Adam, 5,000 steps, five learner seeds). More gates lower the whole envelope (claim (b), reproduced), while within the reachable range the achieved MMD^2 *rises* with target I_{QCLI}, the opposite of claim (a). **(b)** Circuits optimised to maximise I_{QCLI} only, $n = 8, 12, 16$, five seeds and 2 gain settings per n, 3,333 steps each and about 33,000 logged points per n. I_{CCI} never enters the objective.

2×10^{-5}. Claim (a) runs the other way inside the range we can reach, the achieved MMD^2 *rising* as the target's exact I_{QCLI} goes from 0 to 0.41 (Spearman $\rho = +0.979 \pm 0.001 / + 0.901 \pm 0.001 / + 0.809 \pm 0.004$), a trend that also holds for a control trained on exact pmfs. The paper's premise still holds mechanistically ($\rho \leq -0.98$ for the exact-pmf form of the Proposition II.2 floor proxy and ≤ -0.92 for its 10^4-sample form), but the achieved MMD is dominated by trainability rather than support mismatch. Our target family saturates at exact $I_{\mathrm{QCLI}} \approx 0.41$, and at 0–0.064 when estimated from 10^4 samples as the paper does, so the regime claim (a) asserts lies outside what we could build.

D The I_{QCLI}–I_{CCI} ceiling

We also reproduced the experiment in which maximising I_{QCLI} alone raises I_{CCI} as a by-product (Fig. 4(b)). The coupling reproduces at low I_{QCLI} (pooled Spearman $\rho = +0.79 / + 0.79 / + 0.90$ for $n = 8/12/16$ restricted to $I_{\mathrm{QCLI}} \leq 0.3$, and $+0.92 \pm 0.10 / + 0.91 \pm 0.04 / + 0.96 \pm 0.01$ seed by seed) but reverses above $I_{\mathrm{QCLI}} \approx 0.38$ at all three sizes once five seeds are pooled (global $\rho = +0.08, -0.57, -0.64$), the per-seed global ρ itself seed-dependent at the two smaller sizes ($+0.25 \pm 0.92$ and -0.52 ± 0.36) because whether a trajectory reaches the ceiling inside 3,333 steps is a property of the seed. The reversal is forced by the definitions, the exactly uniform distribution maximising I_{QCLI} (0.982/0.998/0.9999) with $I_{\mathrm{CCI}} = 0$. The reference paper's figure stops below $I_{\mathrm{QCLI}} \approx 0.7$, so a high I_{QCLI} need not indicate exploitable structure and can simply indicate proximity to uniform.

E Additional arms

Data scaling. At $N \in \{10, 30, 50, 70, 90, 100\}$ the learned-direction IQP arm is flat for $N \geq 30$ (raw Conv2D-MMD^2 0.068–0.072) and so is the RBM (0.055–0.067), while the DCGAN leads already at N=10 (0.046 ± 0.017 against 0.095 ± 0.002) and falls a further 2× by N=100 ($0.046 \rightarrow 0.022$), as Fig. 5 shows. On the flux-aligned column the IQP arm leads at N=10 and N=30 and the DCGAN from N=50, the two overlapping within a seed sd everywhere except N=10 and N=90.

Exact KL. KL needs the full simulated distribution and cannot be estimated from hardware shots, so we keep it out of Table 2 and record it here at seed 0 only. Its M=2 arm, on 3,094 / 3,272 parameters, reaches PDF–JS 0.00172 at N=11 and 0.00302 at N=100, raw Conv2D-MMD^2 0.119 and 0.091, flux-aligned 0.046 and 0.075. The collapsed M=8 arm of Fig. 2(b) is also a KL arm.

Prior draws at M=2. All draws from the M=2 prior are rings of rms radius 9.40 ± 0.21 px against 9.21 ± 0.15 for real data, though under-dispersed in ellipticity (0.048 against 0.099).

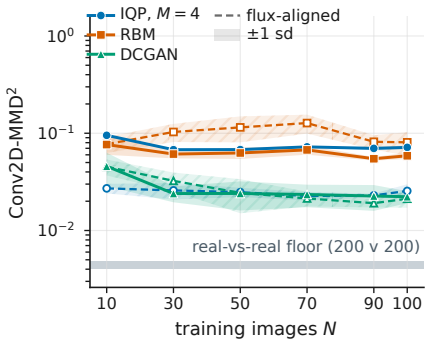


Figure 5: Conv2D-MMD^2 against training-set size, mean $\pm$ sd over five seeds, raw as solid lines and flux-aligned as dashed with open markers, over the real-vs-real floor band. Seeds vary the model only, the training subsets being nested prefixes of one draw.

Other arms. We report 18 conditions, 16 IQP arms and the RBM and DCGAN baselines. The arms outside Table 2 (Hamming-ordered latent block, other objectives and priors, a pseudo-time conditional arm) are scored under the same metrics in the released tables. Trained jointly rather than sequentially, the Hamming-ordered block reaches PDF–JS 0.00178 at $N{=}11$ and 0.00175 at $N{=}100$, consistent with the sequential protocol, not the block, being what fails in Sec. 3.2; we did not run it sequentially. The $M{=}8$ arm collapses (PDF–JS 0.0102, MMD^2 0.241 at $N{=}100$, seed 0).

Table 2 conventions. The † arm runs Algorithm 1 on our low-variance latent block, whose stage 1 trains the phase core and the anchor's own latent block jointly where the reference protocol freezes a random one. Its Params column counts the shared core plus the per-image latent block at each N; the DCGAN entry is its generator alone, its discriminator adding $53{,}497$ training-only parameters. The RBM is likewise a joint model whose visible biases and 40 of its hidden biases are per image, so it holds $10{,}960$ shared weights plus 50 per image and its total also grows with N. Applying the same validation rule to the IQP arms selects 1k–16k steps and changes no Table 2 cell by more than one sd (under the v10 protocol the paper-kernel PDF–JS at $N{=}11$ improves from 0.92 to 0.78×10^{-3}, the one comparison of sixteen that separates); we keep the fixed $4{,}000$-step numbers.

Protocol details. The paper kernel is the three-bandwidth Gaussian MMD of the IQPopt library [16]. Generated pixels outside $[0, 1]$ are clipped before PDF–JS, a clipped mass of at most 6×10^{-3}, and the floor draws come from the $8{,}710$ validation images the held-out split leaves. The kernel is that library's at $\sigma = \{0.989, 1.729, 2.236\}$, mean Pauli weights 2.00, 0.77 and 0.48; the reference paper adopts that kernel but publishes no bandwidths, so the values follow the library's rule. The Conv2D-MMD^2 bandwidth, fixed once on the held-out features and shared by every row of every table, is 0.005547 raw and 0.002558 flux-aligned. Replacing the $1{,}000$-step exact-KL warm start of θ^0 by a flat-MMD warm start on the same mean crop, making every stage shot-estimable, changes no metric by more than one sd at either N. Of the eight restarts of the held-out projection we keep the one correlating best with the target, which a loss-selected choice trails by at most 0.005, and the zero-initialised restart alone reaches 0.899 ± 0.018 at $N{=}11$ and 0.9684 ± 0.0004 at $N{=}100$. The DCGAN is the only arm not bit-reproducible here, cuDNN having no deterministic transposed-convolution kernel, so one configuration at one seed lands up to 57% apart on raw Conv2D-MMD^2 and 74% on PDF–JS, while every RBM and IQP configuration trained twice reproduced exactly.

Baselines at a fixed budget. Table 3 gives the two baselines at the fixed $20{,}000$ steps of our round-3 protocol beside the validation-chosen budget of Table 2; early stopping helps both, by over an order of magnitude for the DCGAN at $N{=}11$.

F Per-class samples

Figure 6 shows what the per-class runs of Table 4 generate, one row per class, from the arm of Table 2 retrained inside each class, and at $N{=}11$ the same three runs reach held-out projection correlations of 0.9227 ± 0.0014, 0.9248 ± 0.0030 and 0.8986 ± 0.0032 against the pooled arm's 0.9102 ± 0.0026.

Table 3: Baselines at a fixed 20,000 steps against their validation-chosen budget, mean ± sd over five seeds, scored under the round-3 convention (bins and encoder bandwidth set per arm), so the two columns compare with each other and not with Table 2.

		PDF–JS ($\times 10^{-3}$)		MMD^2 ($\times 10^{-2}$)	
Arm	N	20k steps	chosen	20k steps	chosen
RBM	11	4.25 ± 1.40	1.59 ± 0.29	8.6 ± 0.8	5.3 ± 0.7
RBM	100	4.27 ± 3.01	1.16 ± 0.23	9.6 ± 2.6	5.4 ± 0.6
DCGAN	11	27.7 ± 33.6	1.11 ± 0.56	67.5 ± 50.3	5.5 ± 1.7
DCGAN	100	9.01 ± 1.22	0.55 ± 0.12	7.2 ± 1.7	2.0 ± 0.3

Table 4: Per-class generation at $N{=}100$ with the arm of Table 2, trained and scored inside one class, against the pooled arm, mean ± sd over five seeds, in the units of Table 2, with sd to two decimals and to four in the projection correlation. Each class has its own floor, given in the text. **Bold** as in Table 2.

Training set	PDF–JS ($\times 10^{-3}$)	MMD^2 ($\times 10^{-2}$)	MMD$^2_{\mathrm{a1}}$ ($\times 10^{-2}$)	proj. corr.
axion only	2.84 ± 0.08	7.11 ± 0.22	2.08 ± 0.23	**0.9718** ± 0.0004
cdm only	2.77 ± 0.24	7.85 ± 0.19	1.84 ± 0.28	0.9688 ± 0.0004
no sub only	2.95 ± 0.32	7.99 ± 0.13	2.94 ± 0.27	0.9701 ± 0.0002
pooled	*2.91* ± 0.28	*7.16* ± 0.29	*2.55* ± 0.44	*0.9684* ± 0.0004

Every row of Table 4 compares with its own real-vs-real floor and not with another class; in that table's units the floors are axion $0.032/0.36/0.31$, cdm $0.034/0.30/0.32$, no sub $0.035/0.45/0.45$ and pooled $0.041/0.48/0.40$, each with an sd of roughly half its mean. The rows look alike because the three classes differ in only a few per cent of the pixel variance, so we claim each class can be generated on its own, not that the model separates them.

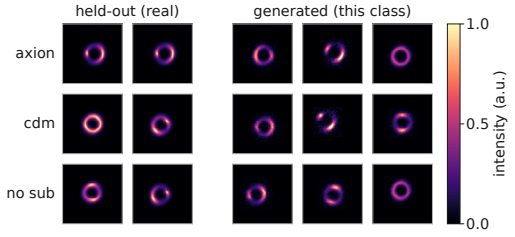


Figure 6: Per-class samples. Each row is one class, two held-out images then three generated by the arm trained on 100 images of that class alone, the first two and first three in index order with no selection, on the shared 0–1 colour scale.